Coding & Solution

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack. 2. The solution should address the core problem statement and implement the key functional requirements. 3. Include all source files, configuration files, and setup instructions needed to run the application. 4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/gummala-sandeep/ai-speech-evaluator
Programming Language(s)	Python 3.11
Framework(s) Used	FastAPI, Streamlit, SQLAlchemy
Key Features Implemented	Whisper STT transcription, Bi-Encoder + Cross-Encoder semantic scoring, Librosa acoustic/filler analysis, weighted scoring engine, ReportLab PDF reports, full authentication and admin CRUD (30/30 planned features, 100% implemented)
Pending / Incomplete Features	None for the MVP scope; future enhancements planned include a Celery async task queue, JWT authentication, and multilingual Whisper support (see Scalability & Future Plan)
Setup / Run Instructions	Clone repo -> create Python 3.11 venv (venv311) -> pip install -r requirements.txt -> configure DATABASE_URL in .env -> bash run.sh (starts FastAPI on :8000 and Streamlit on :8501)

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (GitHub, ~45-60 commits across 4 sprints)

Additional Notes / Comments:

The codebase follows a Domain-Driven Design (DDD) layered architecture (Presentation -> Application -> AI/ML -> Data Access -> Persistence). All 15 code-quality criteria in the accompanying Code-Layout, Readability and Reusability document were assessed as Pass.